

DADA Group project specification

July 8, 2026

Due Monday July 27 at 22:00

1 The task

The group project allows you to explore a topic in some area of computer and information security, in some depth. Your group will present at the showcase on Wednesday 29th July, which will be attended not only by your classmates, but also invited guests. Any supporting documentation/evidence for this presentation is due for submission on Monday 27th July 10pm.

In addition, your group will submit a *short, formal* paper, explaining your topic/area, for assessment. Previous years papers can be found at <https://hugh.myddns.me/media/websites/Proceedings>. Your papers are also due for submission on Monday 27th July 10pm, and will become a course “proceedings”. Your group will become the class experts on your topic. As much as possible, assuming that it appears that you have learnt a lot, and put a lot of effort into your project, you will achieve a good mark.

2 The topic list

The topic specifications given below are deliberately under-specified, and in most cases you can vary your particular topic. You could, for example, choose to demonstrate an attack, or a defence, on exactly the same topic. You could also choose a completely different topic, but this must be agreed to (with Hugh) by Monday 13th

The brief descriptions given below may require further clarification, so please discuss your topic with your lecturer, or Yao Xuan. I have pre-allocated to each group a possible project. If you are group 1, you have been allocated project 1, group 2, project 2 and so on. However, you do not have to choose this topic. If your group, after discussion, wishes to change your topic to another one (even one allocated to another group), then discuss this with Hugh as soon as possible. My advice is that you should meet up as a group, and try to find out the strengths and interests of each of your group members, as soon as possible. You might perhaps consider topics related to Blockchain technology (not necessarily related to BitCoin), forensics, anomaly detection, medical devices,

pentesting, quantum effects, uncopyable keys, NFC man-in-the-middle, hardening phones, false trigger removal from video, malware analysis, APK examiner, Trusted Platform Module (TPM) integration, SIM card sniffing. You might even want to do a more practical project (one with hardware and programming). In any case, your topic selection should be confirmed (by Hugh) on Monday. The initial proposed topics for each group (1.8) in DADA in July 2026 are as follows:

1. **New threat landscapes:** APT, state (and mafia) sponsored computer security defence and attack. You could look at the changing landscape, where AI tools are transforming attack and defence of modern systems.
2. **Hardware, kernel and/or side-channel attacks:** This project would involve exploring “low-level” attacks on machines, or ones that use special purpose hardware (for example attacks on CPU architectures, or attacks using software defined radios and the like), and perhaps developing a demonstration of the attacks. It could involve investigating the security of kernel architectures.
3. **Looking to the future - crypto:** This project would involve exploring emerging crypto technologies. What is on the horizon? Where are we going to be in 10, 20 or 100 years time assuming the current trends in technology? You could go wide, or you could focus on a particular aspect of crypto that interests you (federated learning, or post-quantum techniques, or the like).
4. **Pentesting and/or recovery:** This project would involve exploring pentesting - both the tools that support it, and the techniques to follow when the automated tools give up, and/or digital forensics. Some demonstrations would be nice.
5. **Testing web based systems for vulnerabilities:** Explore how to analyze and/or test (say) PHP for vulnerabilities. This could include techniques for ensuring that systems cannot leak information, or participate in injections, or something else. I am expecting that in this project you would be investigating source code analysis, develop or use some tool, and then reflect on it.
6. **Network defences:** This project would involve exploring the (current) range of network defences, and perhaps looking to the future - what could be done in the future to address emerging threats? Can IPv6 solve our problems? More restrictions?
7. **Looking to the future - authentication:** This project would involve exploring the use of biometric and other methods for authentication. What is on the horizon? Where are we going to be in 10 or 20 years time assuming the current trends in technology? How will you manage your passwords and tokens? What human characteristics can be exploited?
8. **Attacking and defending IoT devices and systems:** An overview of the techniques used to attack and defend IoT devices. It could include discussion on the security architecture of the systems, and/or attack techniques from malicious applications, or via the networks. What is on the horizon? Where are we going to be in 10 or 20 years time?

Other possible topics might include:

1. **Testing *programs* for vulnerabilities:** Explore various techniques used for automatically/formally analyzing programs for vulnerabilities. For example, fuzzing, static analysis. One possibility is you might develop a simulation or visualization of this kind of analysis for teaching purposes. Or perhaps an interface (front end) to an existing tool that demonstrates some aspect of its behaviour.
2. **Security in new applications:** This project would involve exploring unusual requirements and challenges posed by newer applications, for example pay systems or the growing use of self driving cars (or some other tool/app you dream up). For example - with cars, it might be an opportunity for you to imagine a future where cars communicate with each other continuously to evaluate road or traffic conditions, and then to hypothesize what might happen in this scenario. What if there were malicious participants? How can the system protect against this? This might involve software development, or maybe design development.
3. **Testing *systems* for vulnerabilities:** Explore various techniques used for automatically/formally analyzing systems for vulnerabilities. For example, the web servers, or the DNS system, or Android or other operating systems, or the GSM/LTE phone system. All of these, or just one.
4. **Media protection and steganography:** This project would involve exploring the (technical) techniques used in protecting media, digital watermarking and other techniques. One possibility is that you could develop your own digital watermark for (say) images, and then evaluate it's performance against image manipulation (blurring, rotation, resizing, cropping and so on).
5. **Trusted Password Manager built on Intel SGX:** Have you ever wondered why you should trust an online password manager to properly safeguard your credentials? What if a staff member stole your valuable password and misused it? In this project, you will design and develop your own password manager service using Intel SGX, and show off the trustworthiness of your service to users.
6. **AccessPoint/DNS/Wifi spoofing, and/or detector:** This project would involve exploring malicious Access Points, and techniques to defeat them. Hardware to help this may be provided if needed.
7. **State-sponsored Tools and Infrastructure:** This project would involve exploring state sponsored systems (for example Grizzly Steppe and the like).
8. **AccessPoint/DNS/Wifi spoofing, and/or detector:** This project would involve exploring malicious Access Points, and techniques to defeat them.
9. **Multi-stage attacks:** This project would involve exploring multi-stage attacks, perhaps involving some mix of web app, spreadsheets or Word docs, phone, BT, BLE...

10. **BEAST (or similar) attack:** This project is primarily to assist in educating people about attacks like the BEAST attack (or similar M-i-M/encryption attacks). You can develop a simulation, animation, or a visualization of the attack for the purpose of teaching others the import and technique(s) used in the attack(s). Alternatively, perhaps you could develop a test tool for checking if a web site is susceptible to such attacks. Or both.
11. **Ransomware:** This project would involve exploring the (technical) techniques used in ransomware, and the results, with case studies.
12. **BLE/NFC attacks:** This project would involve exploring attacks on BLE or NFC in (for example) Android phones. An example might be DoS, sniffing, or drive-by attack.
13. **New threat landscapes:** Security weakness and possible attacks on blockchain/NFTs. Perhaps a good place to start would be
<https://socradar.io/wp-content/uploads/2024/03/SOCRadar-Cryptocurrency-NFT-Threat-Landscape-Report.pdf>

3 Presentation

The project is to be presented in two forms, firstly as a video+demo presentation, and secondly as a short paper.

Present the project as a formal short paper of 4 to 6 pages in ACM conference format. This link shows a sample formal paper:

<https://hugh.myddns.me/media/websites/Proceedings/ACM/sigproc-sp.pdf>

If you wish, you can use Word - with this sample template file:

<https://hugh.myddns.me/media/websites/Proceedings/ACM/pubform.doc>

You can also use L^AT_EX/LyX (miktex/latex2e), with this class file:

https://hugh.myddns.me/media/websites/Proceedings/ACM/acm_proc_article-sp.cls

You can see some documentation at

<https://www.acm.org/publications/proceedings-template>

You can see some other files for the latex sample at

<https://hugh.myddns.me/media/websites/Proceedings/ACM/>

The *format* of the paper must follow exactly the specified style (including fonts, font sizes, layouts etc). In general, the *structure* of a formal paper should follow that in the sample:

- Title, authors, abstract
- Body of the work - possibly something like this:

- Introduction,
- background knowledge,
- what you did,
- related work, and
- summary/conclusion
- References (and then any appendices)

Appendices can exceed the page limit, if you really cannot reduce your paper to 6 pages. Note: Document (reference) your sources - any *unreferenced* copied text will result in an extraordinarily low mark.

4 Assessment

4.1 Project - expected level?

Your project could end up in various forms, but it should at least have an interesting intuition, topic, or demonstration of something at it's core. YOUR work should comprise a reasonable amount of what you present. Not at the level of original research, but more than just regurgitation. Depending on your project, you may include a demo or github, and perhaps readme.txt files showing how to duplicate your work.

The assessment below indicates how I expect to initially assess the projects. However, the assessment for individual projects may deviate from this in some ways, dependant on the form of the delivered project. Three parts make up the assessment:

- On **Monday 27 July 22:00**, the video and paper are due. They will be assessed by teaching staff with the following (approximate) assessment, which contributes a maximum of **45** marks out of the final (60) project mark.
 - (20) Depth of content: An assessment of the depth of content, and level of effort you have put into the project. The marking schedule will range from 0/20 (if there is almost no evidence of understanding or development of the content; mainly the use of cut-and-paste, and the impression given is: “idle thoughts of idle minds”), through to 20/20 (where there is evidence of excellent understanding or development of the content, ideas are successfully substantiated through sound argument, good use of references, impact and significance is high, clear understanding of solution limits/domains/boundaries and so on).
 - (20) Clarity of content: An assessment of the clarity. The marking schedule will range from 0/20 (if there is almost no evidence of organization of the project idea, the presentation of ideas is poor or not formulated), through to 20/20 (where there is evidence of excellent organization in presentation of project idea, ideas are beautifully and effectively presented and sustained throughout).

- (5) Related work/references: An assessment of the related work, and references.
- On **Wednesday 29th July**, each group will present their topic with a poster. This can be augmented with a short video, and/or you could use PPT/Slides or just talk about your project. Your presentation will be assessed both by teaching staff (10 marks), and via a peer assessment exercise (5 marks). This contributes a maximum of **15** marks out of the final (60) project mark. The assessment will be looking for evidence that you have captured the most important “hard” idea in your topic, and that you have presented this idea clearly.

5 The groups

You should meet up with your team members as soon as possible. Here are the groups, your topic, and your consultation times:

- Group 1: Your (initial proposed) topic is topic 1, and your group members are Cheng Zimu 程自牧, Guo Haotian 郭浩天, Jiang Boxi 姜博曦, Shen Qinmo 申沁墨 and Ye Zedong 叶泽东. Your consultation time with Hugh is from 1:00 to 1:20, every consultation day.
- Group 2: Your (initial proposed) topic is topic 2, and your group members are Li Yulin 李玉麟, Liu Canyu 刘灿宇, Liu Xinjie 刘新杰, Pei Chengfeier 裴程菲儿 and Yang Peize 杨沛泽. Your consultation time with Hugh is from 1:20 to 1:40, every consultation day.
- Group 3: Your (initial proposed) topic is topic 3, and your group members are Li Muqiao 李牧樵, Wu Ruocheng 吴若成, Ye Qingcheng 叶青澄, Zeng Xiangrui 曾祥睿, Zhu Xichen 朱习尘. Your consultation time with Hugh is from 1:40 to 2:00, every consultation day.
- Group 4: Your (initial proposed) topic is topic 4, and your group members are Li Yuwen 李或闻, Liu Wenxu 刘文旭, Ma Tianze 马天泽, Tian Yuhe 田雨禾 and Zhao Guanbo 赵冠博. Your consultation time with Hugh is from 2:00 to 2:20, every consultation day.
- Group 5: Your (initial proposed) topic is topic 5, and your group members are Huang Zi Yuan 黄子元, Li Xingyi 李兴怡, Lou Yanxuan 娄彦轩, Wang Hanxuan 王瀚翊 and Wu Yujin 吴玉瑾. Your consultation time with Hugh is from 2:20 to 2:40, every consultation day.
- Group 6: Your (initial proposed) topic is topic 6, and your group members are Li Xiaocheng 李效成, Lin Yanjun 林彦君, Lin Zhengyi 林钲轶, Liu Yi 刘毅 and Yuan Haolong 袁昊龙. Your consultation time with Hugh is from 2:40 to 3:00, every consultation day.
- Group 7: Your (initial proposed) topic is topic 7, and your group members are Li Ruochuan 李若川, Liu Zeyu 刘泽宇, Shu Jiaqi 舒佳琪, Tang Fuming 唐傅铭 and Zhou Kaihang 周楷航. Your consultation time with Hugh is from 3:00 to 3:20, every consultation day.
- Group 8: Your (initial proposed) topic is topic 8, and your group members are Hu Tianren 胡天任, Huang Zheng 黄正, Shi Zichong 施子翀, Tan Ruijia 谭睿佳 and Zhou Zheyuan 周哲远. Your consultation time with Hugh is from 3:20 to 3:40, every consultation day.

You will meet Hugh on Thursday to discuss your initial ideas, and you should perhaps be discussing your topic immediately, to see if it is going to work with your group members. You can of course change this topic if it is not interesting to your group. If you need any assistance, feel free to contact Hugh at hugh@nus.edu.sg at any time. My room is in COM2, #02-34. I am happy to discuss your projects with you, and suggest ideas, approaches to take, special guidance for your topic and so on.

Please note that the topics are deliberately under-specified, to give you more freedom in choosing particular areas to look at. And to repeat: You can change your topic if you wish!

6 Final notes...

Email the lecturer hugh@nus.edu.sg with your final completed posters and papers (sources, word doc, PPT, latex, PDF...) on or before the due dates.

COOPERATING AND COLLABORATION

You may discuss the problems with your friends, and study any background material with them, but the project *comprises your own group's work*. **Copying** and **cheating** will result in *failing* the project.

In addition, an Internet plagiarism checker will check the project submissions, looking for copying. If you do directly use material from other authors, you should always reference this clearly.

Finally - whatever tools you use, you should provide all the sources you use for these tools - all the scripts and so on. This includes C/python/whatever sources, but also if you use (say) ChatGPT or other AI support tools for some reason. At NUS, representing an AI's output as your own work, without any acknowledgement that that you have used such a tool, is considered plagiarism. I expect that you will make clear statements about your use of AI in your project submission (We used chatGPT to improve the English, and restructure the final part of the paper to better... We used Claude-code to generate test sets for the...).